

Large-Scale and Multi-Structured Databases

Project Design

SkyGraph

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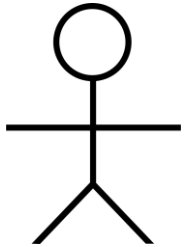
Application Highlights

SkyGraph is an advanced air traffic platform combining real-time flight tracking with historical data analysis.

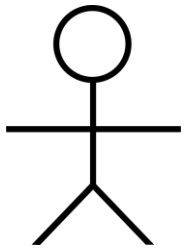
The main features are:

- An integrated system for real-time air traffic visualization and historical flight analysis.
- Interactive map to track active flights.
- Calculation of performance metrics (delays, traffic volumes, distances) on airlines, airports, and specific routes, filterable by time ranges.
- Tools that allow airline companies to evaluate new routes based on shortest path algorithms and hub connectivity ratings.
- Departure/arrival boards, and view of emergency landing for airport flow management.

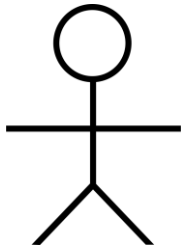
Actors and main mock-ups (i)



Guest (non registered user)

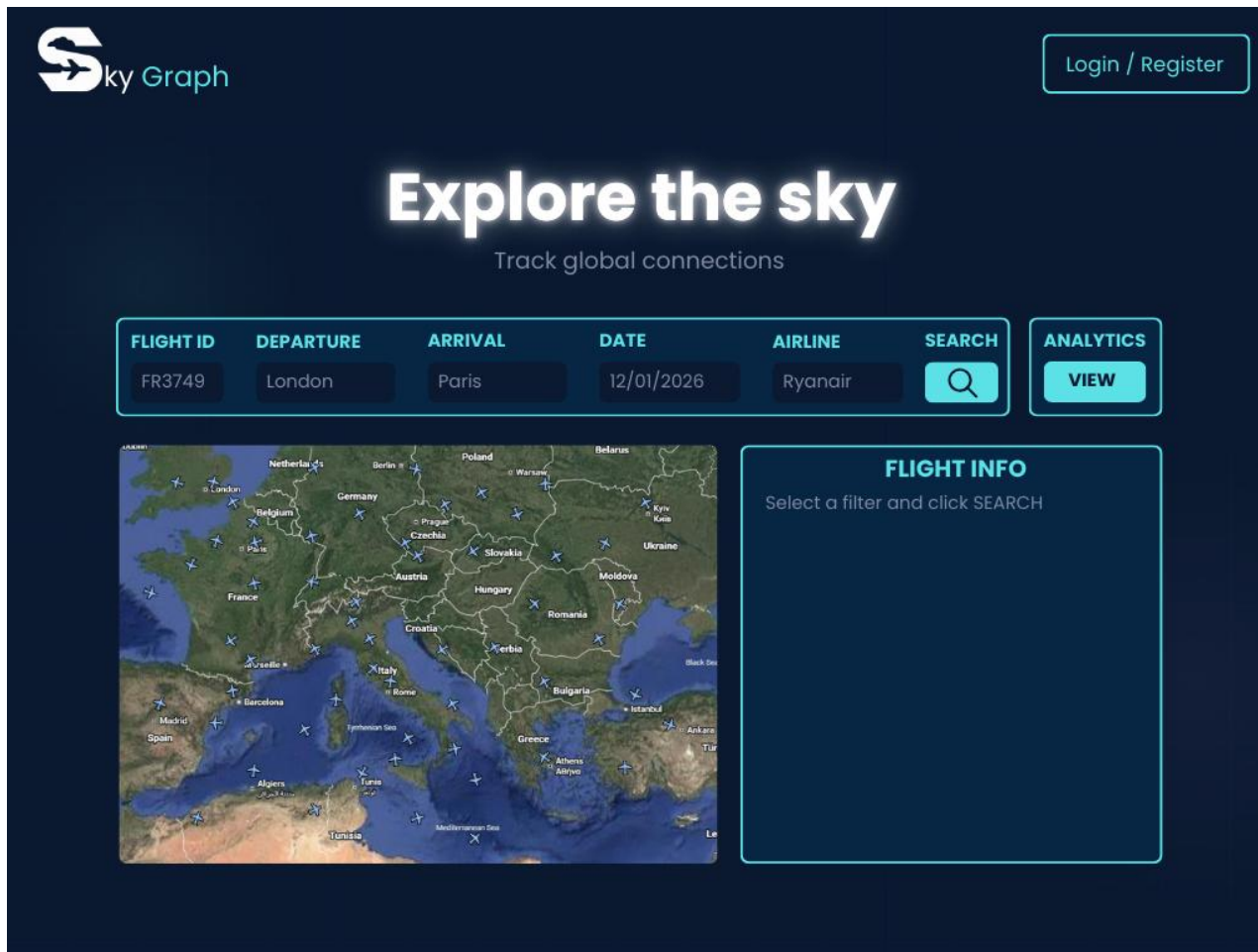


Traffic Controller (registered user)



Airline Representative (registered user)

Actors and main mock-ups (ii)



The main page

Actors and main mock-ups (iii)

Sky Graph Login / Register

Explore the sky

Track global connections

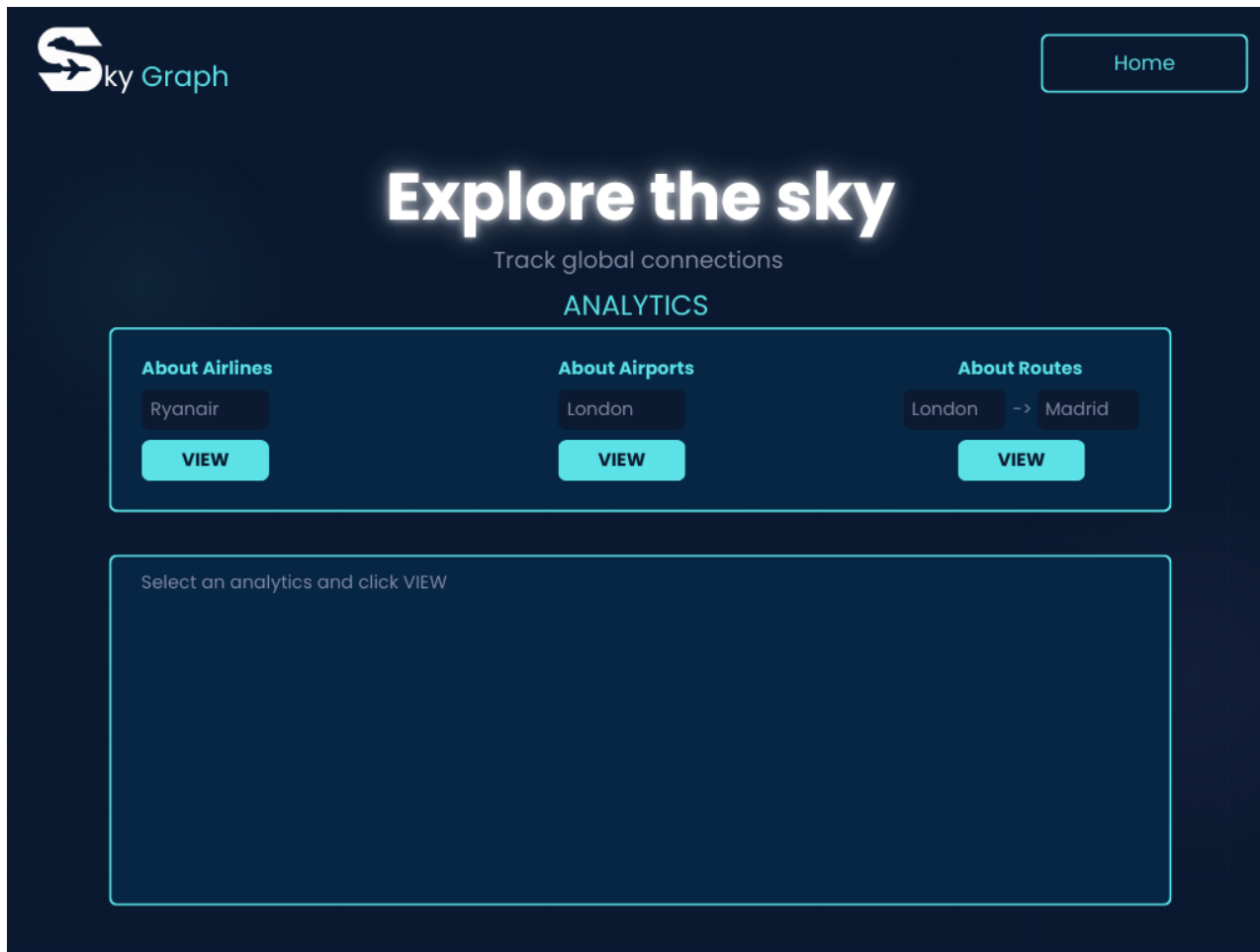
FLIGHT ID	DEPARTURE	ARRIVAL	DATE	AIRLINE	SEARCH	ANALYTICS
FR3749	Pisa	Paris	12/01/2026	Ryanair		VIEW

FLIGHT INFO

- FR1947 | Pisa → Barcellona | WizzAir
07:55 → 09:15 | 12/01/2026 | **Landed**
- BE1942 | Pisa → Dubai | EasyJet
10:15 → 15:15 | 12/01/2026 | **In air**
- BA3712 | Pisa → London | AirWays
10:45 → 12:15 | 12/01/2026 | **Delayed**
- AL1639 | Pisa → Berlin | Ryanair
12:25 → 14:00 | 12/01/2026 | **In air**
- FR3202 | Pisa → Paris | EasyJet
15:50 → 17:00 | 12/01/2026 | **Scheduled**
- BE6725 | Pisa → Milan | Ryanair
17:30 → 18:15 | 12/01/2026 | **Scheduled**
- BA2843 | Pisa → Madrid | WizzAir

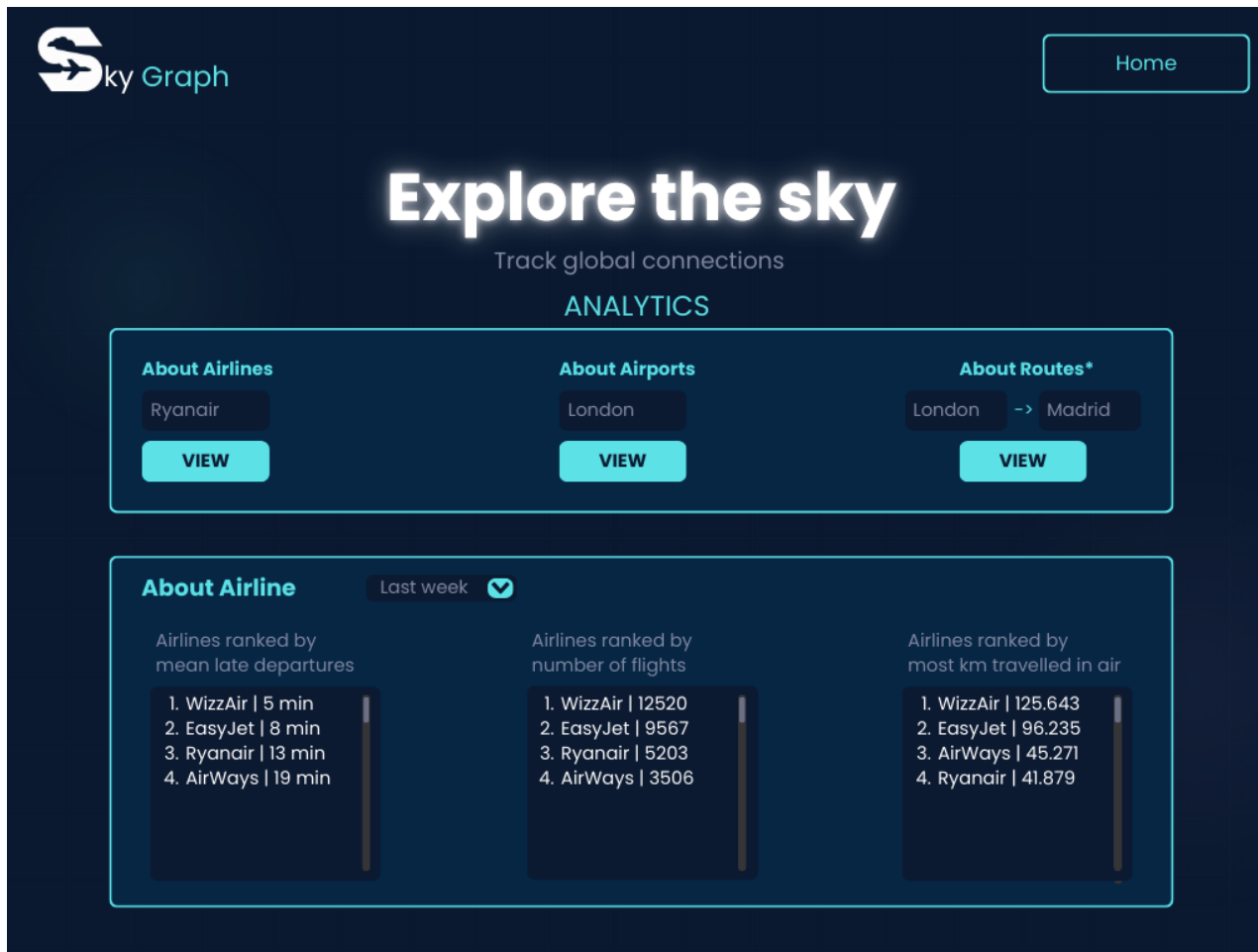
Ex. of a search by some filter inserted

Actors and main mock-ups (iv)



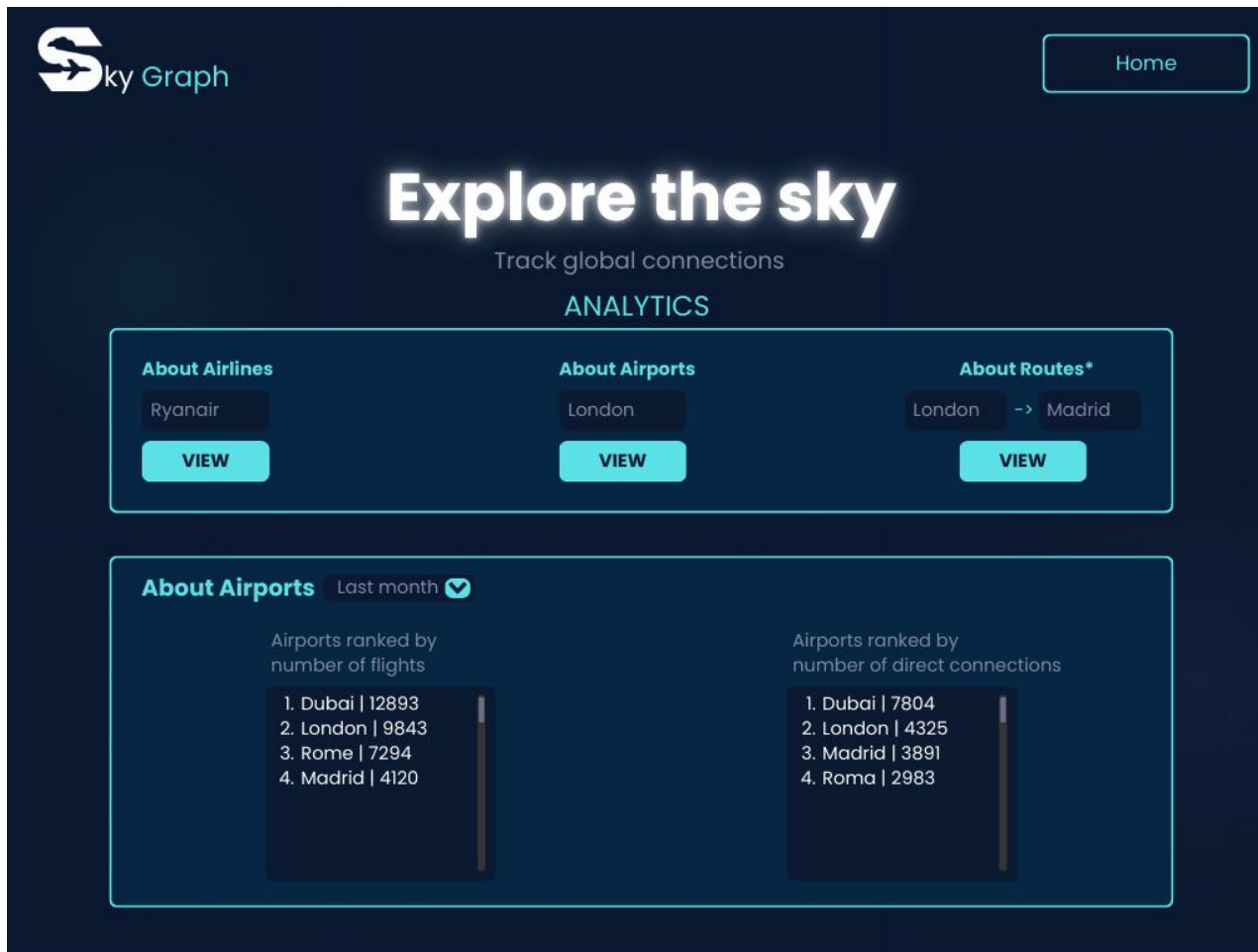
The main analytics page

Actors and main mock-ups (v)



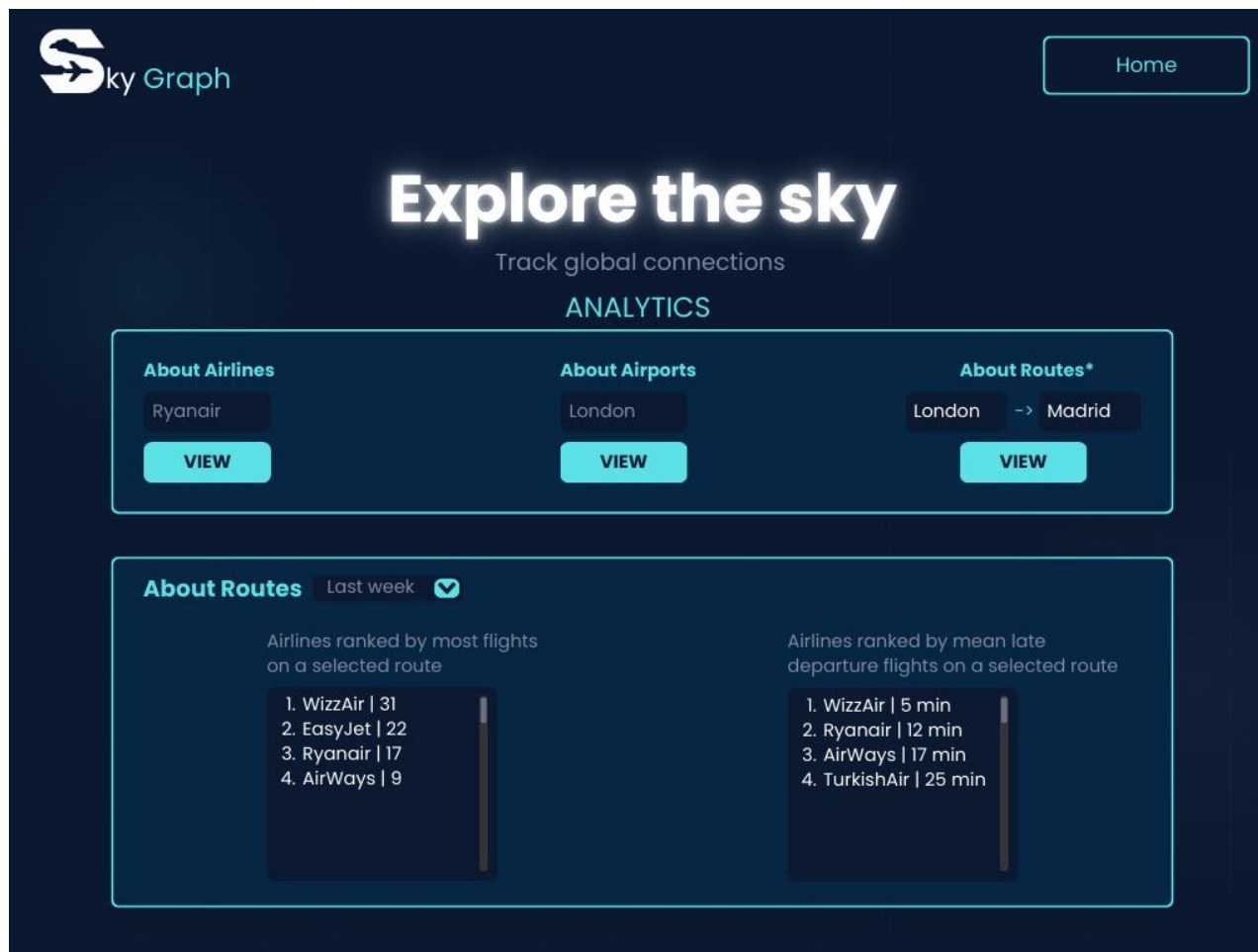
The Airlines analytics

Actors and main mock-ups (vi)



The Airports analytics

Actors and main mock-ups (vii)



The Routes analytics

Actors and main mock-ups (viii)

The mock-up shows a dark-themed web interface for 'Sky Graph'. At the top left is the 'Sky Graph' logo, and at the top right is a 'Home' button. The main heading is 'Explore the sky' with the subtitle 'Track global connections'. Below this is a large container with two sections: 'LOGIN' and 'REGISTRATION'.

LOGIN

E-Mail
JStone@mail.com

Password
...

LOGIN

REGISTRATION

Personal info

First name
John

Second name
Stones

E-Mail
JStone@mail.com

Password
...

About your role

Role
Airline ☒

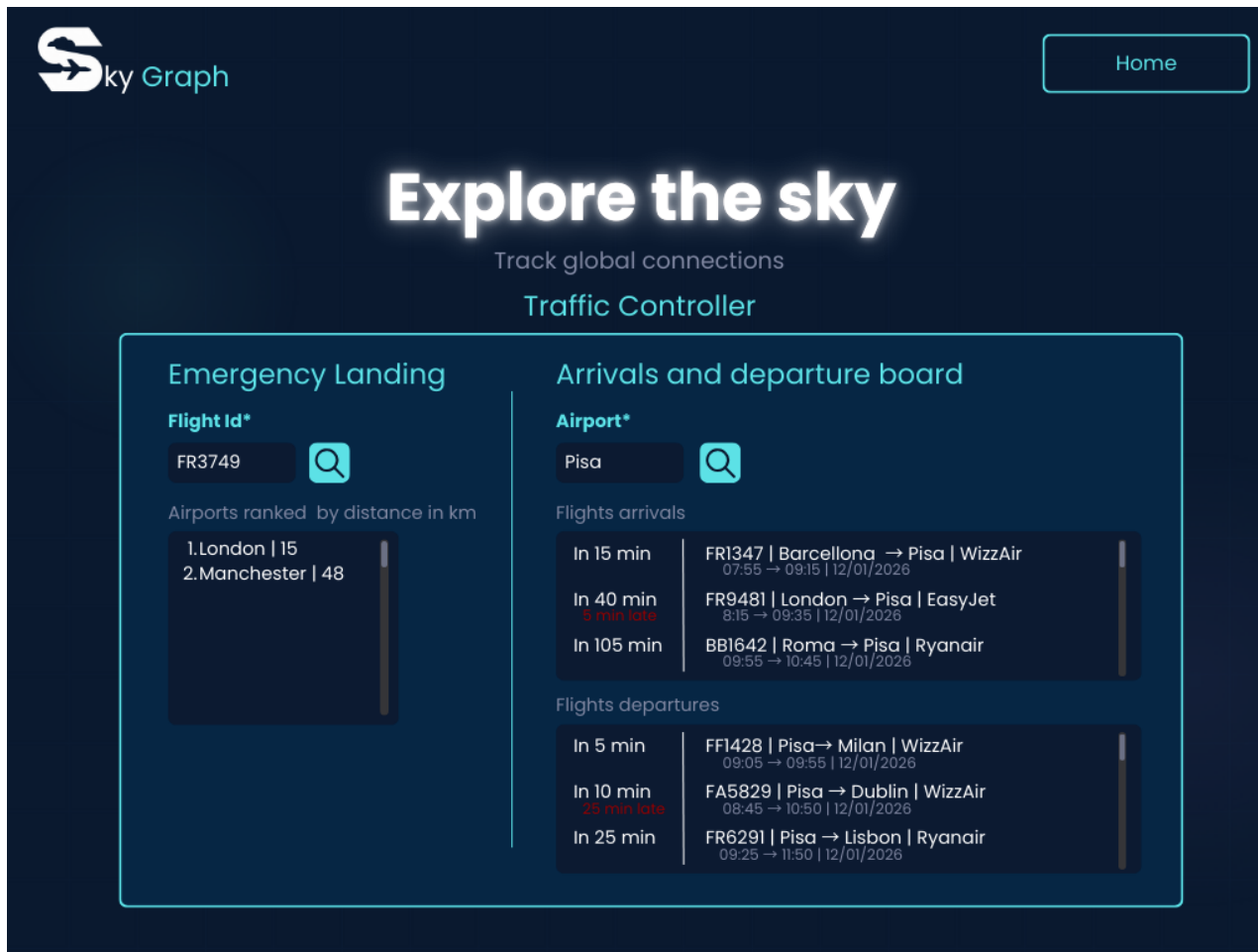
Traffic Controller ID
TC-1234

Airline Representative
AR-1234

REGISTER

The log-in / registration page

Actors and main mock-ups (ix)



The Traffic Controller analytics page

Actors and main mock-ups (x)

The screenshot shows a web application interface for 'Sky Graph'. At the top left is the 'Sky Graph' logo, and at the top right is a 'Home' button. The main heading is 'Explore the sky' with the subtitle 'Track global connections' and the role 'Airline Representative'. The interface is divided into two main sections: 'New Route' and 'View Routes'.

New Route Section:

- From*:** Paris
- To*:** London
- Departure Date*:** 12/02/2026, 08:55
- Arrival Date*:** 12/02/2026, 10:30
- INSERT** button

View Routes Section:

- From*:** Pisa
- To*:** Los Angeles
- Scales*:** 1 (with a dropdown arrow and a search icon)

Time	Airlines	First scale	Second scale
1. 10:40	WizzAir, Ryanair	New York	
2. 12:15	EasyJet	Lisbon	

View network criticality airport score **VIEW**

Airports ranked by score [0-100]

1. Dubai 100
2. London 92
3. Rome 85
4. Madrid 81

The Airline Representative analytics page

Dataset Description

Source: FlighRadar24 API, [Kaggle Dataset](#), AviationStack API

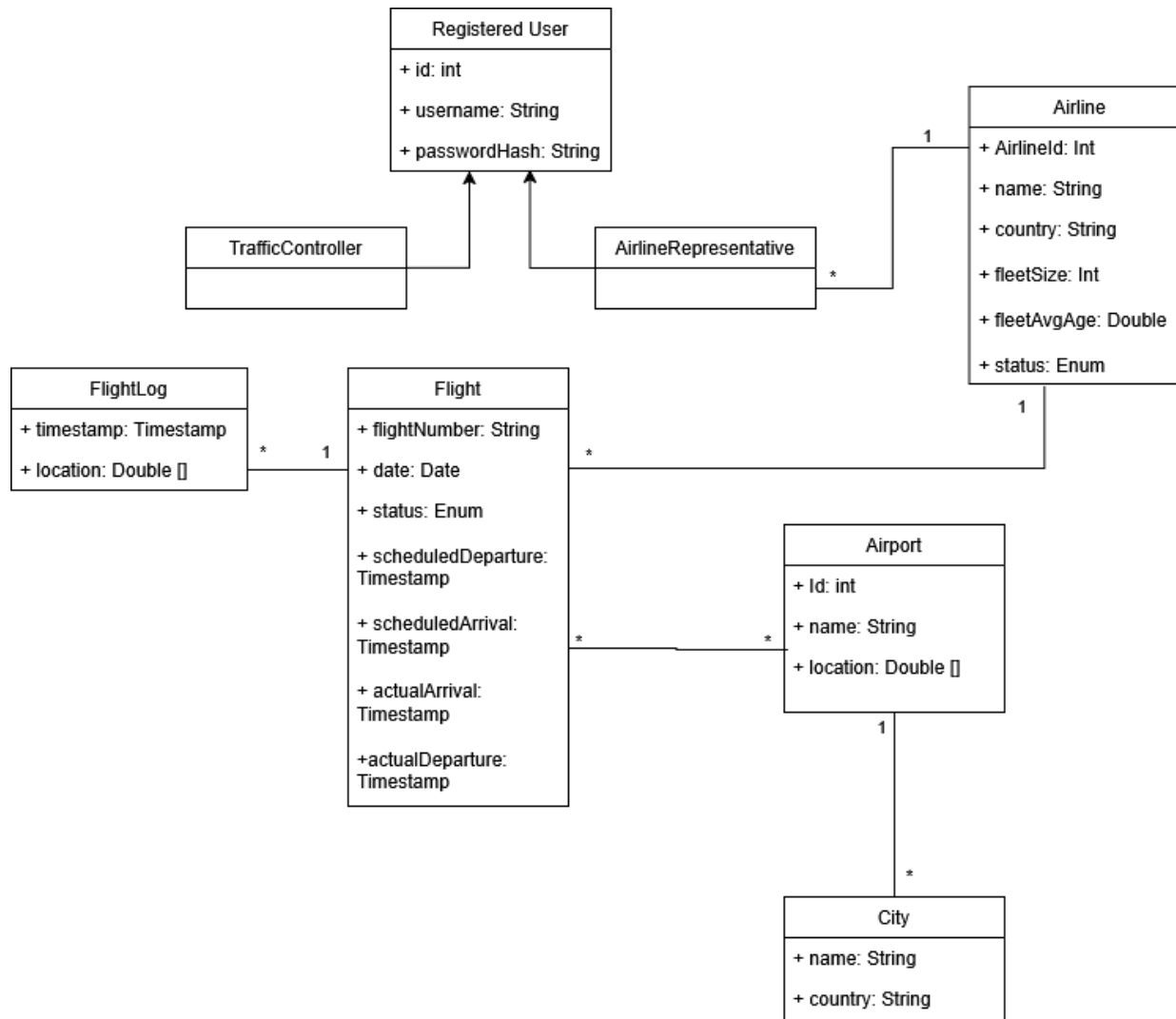
Description: Data concerning flights (FlighRadar24 API), routes (Kaggle Dataset), airports (AviationStack API) and current airplane locations.

Volume: ~600Mb

Variety: The final dataset is obtained by combining the three datasets retrieved from Kaggle and external APIs.

Velocity/Variability: Data about position of Airplanes is quickly outdated, and data about flights loses its relevance as it gets older.

Preliminary UML Class Diagram



Relevant Queries and Entities handled by Document DB

- The system must allow the **Guest** to view aggregate statistics of airlines, airports and routes (in terms of distance travelled, avg delays, number of flights, etc.).
- The system must allow the **Traffic Controller** to identify optimal landing airports in real-time based on proximity and traffic congestion in case of emergency landing.
- The system must allow the **Traffic Controller** to visualize airspace congestion, highlighting the geographic cells with the highest density of aircrafts.
- The system must allow the **Airline Representative** to view, for each airline, the average distance travelled per flight.
- The system must allow the **Airline Representative** to analyze route efficiency via "Reliability" (cancellations) and "Deviation" (fuel waste) indexes.

Entities handled by the DocumentDB:

- Flights
- FlightLog
- Airlines
- Airports
- TrafficController
- AirlineRepresentative

Requirements and Entities handled by Graph DB

- The system must allow the **Traffic Controller** to automate response to airport closures by retrieving affected flights and identifying viable alternative hubs for passenger protection by finding close by open airports with similar flights.
- The system must allow the **Airline Representative** to evaluate potential new routes by searching for complex multi-hop itineraries applying hybrid constraints.
- The system must allow the **Airline Representative** to detect critical infrastructure bottlenecks (e.g., isolated nodes) to prevent network fragmentation.
- The system must allow the **Airline Representative** to detect implicit economic regions by performing Community Detection algorithms on the graph, grouping cities based on high traffic density.

Entities handled by GraphDB:

- Flight
- Airport
- City

